

# ARMAN GHEYSARI

Amirkabir University of Technology (Tehran Polytechnic)

[armangheysari@gmail.com](mailto:armangheysari@gmail.com) · [LinkedIn](#) · [Github](#)

## EDUCATION

NOV 2020- AUG 2023

**MASTER, AMIRKABIR UNIVERSITY OF TECHNOLOGY (TEHRAN POLYTECHNIC)**

Major in computer engineering - Computer Architecture

GPA: 16.43/20

Supervisor: Prof. Hamid Reza Zarandi

**Research Focus:** Fault Tolerance, Distributed Systems, Blockchain, Heuristics Algorithms, Reinforcement Learning

OCT 2014- SEP 2019

**BACHELOR'S, URMIA UNIVERSITY OF TECHNOLOGY**

Major in computer engineering - Information Technology

GPA: 15.91/20

## RESEARCH INTRESTS

- Reinforcement Learning
- Robotics
- Machine Learning
- Deep Learning
- Fault Tolerance
- Fault-Tolerant Protocols
- Distributed Systems
- Blockchain
- Consensus Algorithm
- Operating Systems

## PUBLICATION

ELSEVIER, OCT 2025

**SECURITY-AWARE OPTIMIZATION OF POW-BASED BLOCKCHAIN PERFORMANCE USING AGENETIC ALGORITHM APPROACH **

- Designed a GA-based approach to optimize PoW blockchain parameters: block size, block interval, and propagation mechanism.
- Balanced security and performance, ensuring resilience against selfish mining and Eclipse attacks.
- Reduced median block propagation time and stale block rate, without compromising attack resistance.

- Applied simulation-based evaluation, iteratively refining configurations with GA for optimal security-performance trade-offs.

**SPRINGER, AUG 2025**

### **SIMULTANEOUS OPTIMIZATION OF NETWORK-ON-CHIP TO IMPROVE RELIABILITY AND REDUCE AVERAGE PACKET LATENCY CONSIDERING BUFFER SIZE CONSTRAINTS**

- Proposed a multi-objective optimization framework (NSGA-II) for mesh-based NoCs with 5-port routers and XY routing algorithm. Used buffer size as the main optimization parameter impacting latency, cost, and reliability.
- Validated with Garnet 2.0 simulator; tested networks from 3×3 to 8×8 under uniform, hotspot, and real traffic workloads.
- On a 4×4 mesh (uniform traffic): achieved 18.0001 cycles average latency and reduced failure rate to 4213 FIT (failures per billion hours).

**IEEE, APR 2025**

### **OPTIMIZING PROOF-OF-WORK BLOCKCHAIN NETWORK COMMUNICATIONS USING GENETIC ALGORITHM**

- Introduced a GA-based topology optimization method for Proof-of-Work (PoW) blockchain networks.
- Optimized node placement and interconnections to improve network communication efficiency.
- Achieved a 20.2% reduction in communication overhead during transaction propagation and block dissemination.

## **PATENTS**

**PATENT NO. 140250140003006003 · ISSUED NOV 2023**

### **METHOD FOR MITIGATION AND DETECTION OF CRASH AND BYZANTINE FAULTS IN SYSTEM-ON-CHIPS USING BLOCKCHAIN & CONSENSUS ALGORITHMS**

- Developed a fault detection and mitigation system for safety-critical SoCs.
- Incorporated PBFT consensus modules and per-core blockchain storage to ensure correctness and tamper-proof execution.
- Used Merkle tree root verification, a novel arbitration circuit, and a custom algorithm to isolate faulty processing units.
- Improved system reliability and energy efficiency by removing compromised cores from computation cycles.

**PATENT NO. PCT/IB2024/061908 · FILED NOV 2024**

### **FAULT-TOLERANT SYSTEM-ON-CHIP**

- Proposed a Byzantine fault-tolerant SoC architecture for many-core processors.
- Each processing element (PE) is paired with a consensus module that validates SOC controller instructions using PBFT.
- Execution results are stored in blockchain logs, and Merkle tree roots are sent to an arbiter module.

- The arbiter identifies faulty PEs and informs the SOC controller to deactivate compromised elements.
- Ensures high system dependability by dynamically purging and reconfiguring around faulty units.

## IN-PROGRESS RESEARCH

- Safe Offline-to-Online Reinforcement Learning for Fault-Tolerant Distributed Systems

## TEACHING EXPERIENCE

### SPRING 2025

#### **FAULT TOLERANCE**, TEACHER ASSISTANT

Presented by: Prof. Hamid Reza Zarandi

### SPRING 2023

#### **TESTING & TESTABLE DESIGN**, TEACHER ASSISTANT

Presented by: Prof. Hamid Reza Zarandi

### SPRING 2021

#### **OPERATING SYSTEMS**, LAB INSTRUCTOR

C/C++/PYTHON/UBUNTU

### FALL 2018

#### **ROBOTIC TRAINING COURSE**, INSTRUCTOR

PYTHON/LINUX/RASPBERRY PI

## WORK EXPERIENCE

- Security & Data Analyst - Amirkabir University of Technology (Tehran Polytechnic) [JUL 2024 - Present]
- INTERSHIP - THE CENTRAL TELECOMMUNICATIONS COMPANY, WEST AZERBAIJAN, IRAN [JUN 2017 - SEP 2017]

## RESEARCH EXPERIENCE & FREELANCE

- Implementing a Blockchain Using PoW Consensus Algorithm [Aug 2023]
- Implementing a Blockchain Using PBFT Voting-Based Consensus Algorithm [Dec 2022]
- Implemented a 64-bit UNIX like operating system from scratch [OCT 2021]
- Implementing Parking System Using FPGA and HW/SW Co-design in vivado [SEP 2021]
- Developing Simple Compiled-Code Compiler with Python [JAN 2021]
- Developing Simple Event-Driven Compiler with Python [JAN 2021]
- Developing Four OWASP-Nettacker Plugin for Bachelor's Project Using Python [FEB 2019]
- Developing Path Planning Algorithm and Embed in ROS for Rescue Robot [OCT 2018]
- Developing Object Detection Using OpenCV Library, YOLO Algorithm and Python [OCT 2018]
- Developing 3D-Map Generator using Microsoft Kinect Sensor and ROS [NOV 2018]

- Developing Python Framework for Controlling, Motors, Servo Motors & Actuators [NOV 2018]
- Developing Python Scripts for Reading Data from Various Sensors [NOV 2018]
- Developing Routing Algorithm Using Genetic Algorithm for AI Course [DEC 2018]
- Implementing A simple C Compiler for Synthesis C code Using C++ [JAN 2016]
- Developing a CMS Using Php&Mysql as Internship Project [SEP 2017]

## SKILLS

- Programming
  - C/C++ Programming
  - Python Programming
  - Perl Programming
  - Bash Script Programming
  - HDL SystemC Programming
  - Network-DataLog (NDLog) Programming - Declarative Network Language Programming
- System
  - CCNA Routing&Switching
  - Linux Professional Institute LPIC-1
  - Linux Professional Institute LPIC-2
  - ROS(Robotic Operating System)
  - Advanced Penetration Testing With Kali Linux
  - Information Security Management System Auditor (ISMS) (ISO 19011:2011 ISO/IEC 27001:2013)
- Software & Simulator
  - GEM5
  - Cisco Packet Tracer
  - Burp Suite
  - Wireshark
  - NS-3 Network Simulator
  - BFTSim Simulator
  - RapidNet Simulator
  - Noxim
  - Modelsim
  - Git/Github
- Library & Toolkits
  - PyTorch
  - Tensorflow
  - Keras
  - Numpy
  - Pandas
  - Scikit-Learn
  - Seaborn
  - Matplotlib
  - Optuna
  - Scapy

## HONORS AND AWARDS

- Ranked 3<sup>st</sup> Among 13 Peer M.Sc. Student in Computer engineering – Computer Architecture
- Ranked 190<sup>th</sup> among 20000-25000 Participants In Nation-Wide University Entrance of Computer Eng.
- Member of Urmia University of Technology Robotics Team as Head of the Python Programming Team

## CERTIFICATES

- Fundamentals of Reinforcement Learning
- Modern Robotics: Mechanics, Planning, and Control Specialization - Coursera
- Network+ Certified - Sharif University of Technology (LAITEC) [FEB 2016]
- CCNA R&S Certified - Cando Training Center [APR 2016]
- LPIC-1 Certified - Linux Professional Institute (lpi.org) [DEC 2016]
- ISMS Certified – Batis Academy [JAN 2017]

## LANGUAGES

- Persian: Native
- English: Second Language

## NOTABLE COURSES

- Advanced Operating System: 19.32
- Advanced Computer Architecture: 19.1
- Testing&Testable Design: 18.21
- Basics of AI and Applied Artificial Intelligence: 17.50
- Fault Tolerance: 16.35